

## ROBERTSON LAKE, NU, Canada

WMO: 714900

Lat: 65.100N

Lon: 102.430W

Elev: 800

StdP: 14.28

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -43.8      | -41.3 | -51.0                         | 0.3 | -43.4 | -48.0 | 0.3 | -40.7 | 37.8                  | -11.7 | 35.7 | -11.2 | 10.4                  | 260  | 1.088 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.5                   | 74.9            | 57.2 | 70.1 | 56.3 | 65.2 | 54.5 | 60.0                | 70.6 | 57.7 | 66.8 | 55.2 | 63.3 | 13.8                 | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.5                            | 67.4 | 62.7 | 53.1 | 61.9 | 61.0 | 50.9 | 56.8 | 58.6 | 26.8          | 70.5 | 25.2 | 66.6 | 23.6 | 63.5 | 75.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 35.7              | 30.8 | 27.4 | -46.0                      | 81.1 | 3.3                | 4.6 | -48.4                                              | 84.4 | -50.3      | 87.1 | -52.2      | 89.7 | -54.6      | 93.0 |  |  |
|                   |      |      | -45.7                      | 64.6 | 3.6                | 4.4 | -48.3                                              | 67.8 | -50.4      | 70.4 | -52.5      | 72.8 | -55.1      | 76.0 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 12.4        | -23.5 | -22.5 | -14.1 | 2.3   | 19.9  | 43.1  | 53.6  | 49.6  | 37.1  | 18.7  | -2.1  | -16.2 | (6)  |
|                                                                        | DBStd       | 29.39       | 11.76 | 11.98 | 13.21 | 12.58 | 10.05 | 8.46  | 7.10  | 6.57  | 7.99  | 9.79  | 11.22 | 12.00 | (7)  |
|                                                                        | HDD50       | 14008       | 2278  | 2030  | 1987  | 1431  | 933   | 240   | 43    | 88    | 393   | 969   | 1564  | 2052  | (8)  |
|                                                                        | HDD65       | 19220       | 2743  | 2450  | 2452  | 1881  | 1398  | 656   | 358   | 480   | 837   | 1434  | 2014  | 2517  | (9)  |
|                                                                        | CDD50       | 269         | 0     | 0     | 0     | 0     | 0     | 35    | 154   | 74    | 6     | 0     | 0     | 0     | (10) |
|                                                                        | CDD65       | 6           | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 1     | 0     | 0     | 0     | 0     | (11) |
|                                                                        | CDH74       | 146         | 0     | 0     | 0     | 0     | 0     | 25    | 101   | 20    | 0     | 0     | 0     | 0     | (12) |
|                                                                        | CDH80       | 16          | 0     | 0     | 0     | 0     | 0     | 4     | 9     | 3     | 0     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 13.7        | 14.0  | 13.9  | 13.9  | 13.8  | 13.4  | 11.7  | 12.5  | 13.6  | 15.0  | 15.5  | 13.8  | 13.9  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 9.1         | 0.3   | 0.2   | 0.4   | 0.4   | 0.5   | 0.8   | 1.5   | 1.6   | 1.4   | 1.0   | 0.5   | 0.4   | (15) |
|                                                                        | PrecMax     | 11.1        | 0.8   | 0.7   | 1.1   | 1.0   | 1.1   | 1.6   | 2.6   | 2.6   | 2.4   | 1.7   | 1.0   | 0.7   | (16) |
|                                                                        | PrecMin     | 6.1         | 0.0   | 0.0   | 0.1   | 0.1   | 0.1   | 0.3   | 0.6   | 0.7   | 0.7   | 0.2   | 0.1   | 0.0   | (17) |
|                                                                        | PrecStd     | 1.2         | 0.2   | 0.2   | 0.2   | 0.2   | 0.2   | 0.4   | 0.5   | 0.5   | 0.5   | 0.4   | 0.2   | 0.2   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 17.6  | 10.8  | 21.2  | 32.6  | 42.8  | 77.3  | 81.4  | 76.3  | 64.2  | 40.8  | 27.3  | 19.2  | (19) |
|                                                                        |             | MCWB        | 17.5  | 10.4  | 20.8  | 31.8  | 39.3  | 58.0  | 60.3  | 57.4  | 55.2  | 39.3  | 27.2  | 18.5  | (20) |
|                                                                        | 2%          | DB          | 7.2   | 5.7   | 16.1  | 28.4  | 38.5  | 69.3  | 77.1  | 69.9  | 57.7  | 36.2  | 23.0  | 13.3  | (21) |
|                                                                        |             | MCWB        | 6.7   | 5.2   | 15.7  | 27.3  | 36.4  | 55.7  | 57.9  | 55.4  | 51.1  | 35.1  | 22.7  | 12.9  | (22) |
|                                                                        | 5%          | DB          | 0.0   | 2.1   | 11.0  | 25.0  | 36.0  | 63.0  | 73.1  | 65.3  | 52.8  | 33.8  | 19.2  | 7.7   | (23) |
|                                                                        |             | MCWB        | -0.5  | 1.6   | 10.6  | 24.4  | 34.5  | 53.0  | 57.4  | 54.3  | 48.0  | 33.2  | 18.8  | 7.3   | (24) |
|                                                                        | 10%         | DB          | -6.1  | -2.0  | 6.6   | 21.1  | 33.5  | 58.0  | 68.5  | 61.1  | 48.2  | 31.5  | 15.7  | 2.6   | (25) |
|                                                                        |             | MCWB        | -6.4  | -2.4  | 6.2   | 20.5  | 32.5  | 50.0  | 55.9  | 52.1  | 45.6  | 30.9  | 15.3  | 2.2   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 15.1  | 10.7  | 21.2  | 32.2  | 40.2  | 61.2  | 63.7  | 60.9  | 56.2  | 39.9  | 27.4  | 18.1  | (27) |
|                                                                        |             | MCDB        | 15.2  | 11.0  | 21.4  | 32.7  | 42.2  | 72.0  | 76.8  | 71.2  | 63.0  | 40.5  | 27.5  | 18.4  | (28) |
|                                                                        | 2%          | WB          | 5.8   | 5.5   | 16.2  | 27.5  | 36.5  | 57.1  | 61.1  | 57.9  | 51.8  | 35.6  | 23.3  | 11.0  | (29) |
|                                                                        |             | MCDB        | 6.2   | 5.9   | 16.5  | 28.2  | 37.9  | 67.8  | 72.2  | 66.2  | 57.1  | 36.1  | 23.5  | 11.4  | (30) |
|                                                                        | 5%          | WB          | -1.0  | 2.1   | 11.4  | 24.5  | 34.4  | 54.0  | 59.1  | 55.4  | 48.7  | 33.2  | 19.3  | 6.8   | (31) |
|                                                                        |             | MCDB        | -0.5  | 2.5   | 11.7  | 25.2  | 35.6  | 63.4  | 69.3  | 62.3  | 52.3  | 33.8  | 19.5  | 7.3   | (32) |
|                                                                        | 10%         | WB          | -6.8  | -1.7  | 6.9   | 20.5  | 32.4  | 50.9  | 56.8  | 53.2  | 45.9  | 31.0  | 15.8  | 1.5   | (33) |
|                                                                        |             | MCDB        | -6.4  | -1.3  | 7.2   | 21.2  | 33.2  | 57.9  | 65.5  | 59.5  | 48.4  | 31.5  | 16.0  | 1.9   | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 11.9  | 12.6  | 14.4  | 15.9  | 12.3  | 17.2  | 18.5  | 15.2  | 10.8  | 9.5   | 12.0  | 12.3  | (35) |
|                                                                        |             | MCDBR       | 18.9  | 17.2  | 15.6  | 17.4  | 12.6  | 27.9  | 27.4  | 23.7  | 19.8  | 9.4   | 14.0  | 17.1  | (36) |
|                                                                        | 5% WB       | MCWBR       | 19.0  | 16.3  | 15.7  | 16.9  | 10.9  | 17.2  | 12.3  | 12.5  | 13.2  | 8.9   | 14.2  | 17.3  | (37) |
|                                                                        |             | MCWBR       | 18.7  | 16.4  | 15.6  | 17.8  | 12.0  | 26.5  | 23.8  | 19.9  | 18.1  | 9.6   | 13.8  | 17.3  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.146 | 0.227 | 0.246 | 0.282 | 0.302 | 0.328 | 0.345 | 0.316 | 0.297 | 0.241 | 0.157 | 0.142 | (40) |
|                                                                        |             | taud        | 1.911 | 2.139 | 2.256 | 2.193 | 2.245 | 2.453 | 2.441 | 2.501 | 2.520 | 2.279 | 1.914 | 1.926 | (41) |
|                                                                        | Ebn at Noon |             | 134   | 218   | 271   | 282   | 286   | 281   | 272   | 271   | 253   | 211   | 116   | 46    | (42) |
|                                                                        |             | Edh at Noon | 7     | 17    | 25    | 34    | 36    | 30    | 30    | 26    | 20    | 15    | 6     | 2     | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 49    | 292   | 873   | 1656  | 2091  | 1802  | 1651  | 1185  | 650   | 207   | 74    | 14    | (44) |
|                                                                        | RadStd      |             | 2     | 9     | 37    | 55    | 133   | 133   | 138   | 115   | 81    | 44    | 9     | 0     | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page